

The hardware & software that hold a network together

Every cyber security incident in this course happens on a network. Before you can defend one, you need to know the gear inside it — what each box does, and which ones are actually there to keep attackers out.

What this key knowledge covers

This summary distils everything you need for **KK02 — The hardware & software that hold a network together** in Cyber Security. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

You can't protect a network you can't picture

This dot point is the **vocabulary and the map** for the whole Cyber Security outcome. The VCAA wording is short — "*key hardware and software components of networks*" — but it underpins everything that follows: threats (KK06–KK08), risk reduction (KK09), encryption (KK10) and the legislation (KK13). In the SAC you'll be handed a case study about a real-ish organisation's network and asked to **name components, explain what they do, and say which ones reduce risk**. If you mix up a router and a switch, or a firewall and antivirus, you lose easy marks.

Clutch is a Melbourne-built learner-driver app: it matches L-platers with driving instructors, logs the 120 supervised hours, and processes lesson payments. It holds **permit numbers, home pick-up addresses, parent contacts, payment cards and live GPS** for roughly 40,000 16–18-year-olds. Clutch runs a small office network plus cloud servers — the perfect place to spot every component in this dot point doing its job. We'll keep using Clutch through KK02–KK13.

Quick orientation: components in a network fall into two buckets the examiner cares about — **things that move traffic** (routers, switches) and **things that protect data** (firewalls, security software, IDS/IPS, backups). Keep that split in your head and most questions sort themselves out.

Meet the gear inside Clutch HQ

Here are the six components named in the study design. **Tap any card to flip it** for the one-line job, a plain-English analogy, and where it sits at Clutch. We'll then take each in turn.

A **post office sorting hub between suburbs**. It reads the destination on each parcel and sends it down the right road toward another network.

An **office receptionist who knows every desk**. A parcel for desk 14 goes straight to desk 14 — not shouted across the whole floor.

A **nightclub bouncer with a guest list**. On the list, you're in; not on it, you're turned away at the door.

Storage is your **working filing cabinet**; a backup is the **photocopy kept off-site in a safe**. Lose the original, you still have the copy.

A **personal bodyguard for each device**. The firewall guards the building's front gate; this guards the individual people inside.

IDS = a **security camera + alarm** (it tells you). IPS = a **guard who slams the gate** (it acts). One watches, one stops.

Think of Clutch HQ as a **house**. The **router** is the front door to the street (other networks). The **switch** is the hallway connecting the rooms inside. The **firewall** is the security gate at the boundary. **Storage & backup** is the safe (and the copy at grandma's house). **Security software** is each person's own caution. **IDS/IPS** is the camera-and-guard combo watching for intruders.

Routers vs switches — the #1 thing students confuse

Both move data, so they look similar — but they work at different levels and the examiner loves to test the difference. The one-line rule:

A **switch connects devices *within* a single network** (a LAN). A **router connects *different* networks together** (e.g. your LAN to the internet). Switch = inside. Router = between.

Router — the path-finder

A router sits at the boundary of a network and decides where traffic goes *next*. It reads the destination **IP address** on each packet and forwards it toward the right network, choosing an efficient path. Home/office routers usually also run **DHCP** (handing out IP addresses to devices) and **NAT** (letting many devices share one public address). Because all traffic to and from the internet passes through it, the router is a natural place to attach a firewall.

Switch — the precise deliverer

A switch connects the devices on one LAN — PCs, printers, servers — and forwards each **frame** only to the port of the device it's addressed to, using **MAC addresses** it has learned. That precision is why a switch beats an older **hub**, which blindly copied every frame to every port (noisy, slow and a security risk, because everyone could see everyone's traffic).

Firewalls — the gatekeeper with a rulebook

A firewall **inspects traffic crossing a boundary** and decides to **allow or block** it based on rules. Rules can match the source/destination IP address, the **port** (which service is being reached), and the protocol. Default-deny is the safe posture: block everything, then open only what's needed.

A **stateful** firewall also remembers active connections, so it can tell the difference between a reply you asked for and an unsolicited probe from a stranger. That's how Clutch's firewall lets staff browse out to the web, while silently dropping random connection attempts coming *in* from the internet toward the payment server.

A **hardware firewall** is a dedicated device (often built into the router) guarding the whole network at its edge — one checkpoint for everyone. A **software firewall** runs on an individual computer (e.g. the one built into Windows), protecting just that device. Most organisations use **both**: perimeter device + per-machine software. (Note the overlap — a software firewall is also a form of "security software" in the next section.)

Storage & backup — survive the worst day

Storage is where data normally lives: file servers, a **NAS** (network-attached storage box) on the LAN, or cloud storage. **Backup** is a *separate, recoverable copy* kept so data survives a disaster — hardware failure, theft, accidental deletion or a **ransomware** attack that encrypts the originals. The distinction is the whole point of the dot point: a second live copy on the same server is *storage*, not a backup, because the same incident can take out both.

The 3-2-1 rule — build it below

The industry shorthand for a backup that actually saves you: keep **3** copies of the data, on **2** different types of media, with **1** copy kept off-site (or offline). Toggle the pieces and then hit a ransomware strike to see what survives.

Imagine Clutch keeps its only backup on a second drive in the same office. A burst pipe, a fire, a theft — or ransomware spreading across the LAN — takes the original *and* the copy. The off-site (or offline) copy is the one the attacker and the disaster can't reach. That's the copy that gets the company back online without paying a ransom.

Security software — protection that runs on the device

Not every defence is a box in a rack. **Security software** is the protective programs running on each computer and server. The examinable families:

- **Anti-malware / antivirus** — scans files and running programs for viruses, worms, spyware and ransomware, and quarantines or removes them.
- **Encryption software** — scrambles stored data (e.g. the lesson database) and traffic so a thief who copies the files still can't read them.
- **Patch & update management** — keeps the operating system and applications current, closing the known holes attackers rely on (links straight to KK09).
- **Endpoint monitoring / software firewall** — watches each device's behaviour and filters its traffic locally.

Hardware or software? A quick gut-check: if you could *uninstall* it, it's security software; if you could *trip over it in the server room*, it's hardware. (A software firewall lives in both worlds — it's software that does a firewall's job.)

IDS vs IPS — watch, or stop?

This pairing is the single most tested distinction in the dot point. Both inspect traffic for signs of an attack — using **signatures** (known attack patterns) and/or **anomaly detection** (anything that strays from normal behaviour). The difference is what they do when they find something.

- Detects suspicious activity and raises an **alert / log** for the security team.
- Usually sits **out of line** (monitoring a copy of traffic) — it watches, it doesn't touch.
- The malicious packet **still gets through**; a human has to respond.
- Detects *and* **automatically blocks / drops** the malicious traffic.
- Sits **inline** (traffic flows through it) so it can stop attacks in real time.
- The malicious packet is **stopped before it lands** — no human needed for the block.

You can prove this to yourself in the lab below — send a stealth attack with only the IDS on and watch it reach the server *with* an alert, then switch the IPS on and watch the same attack get dropped.

Defence-in-Depth Lab

This is the whole dot point in one moving picture. Traffic flows left → right through Clutch's network: **Internet → Router → Firewall → IDS/IPS sensor → Switch → Server** (with a backup branch). Toggle each defence, then fire different packets and watch *which layer catches what*. Notice how no single box stops everything — that's **defence in depth**.

toggle defences · fire packets

The firewall stops obvious, rule-breaking traffic at the edge. A stealth/zero-day attack that *looks* legitimate sails straight past it — that's exactly why you also need an **IPS to block** what the firewall missed, an **IDS to alert** on what gets through, security software on the endpoint, and a clean **backup** for when something still lands. Layers cover each other's gaps.

Quick quiz — 6 questions

Multiple choice, instant feedback. These mirror the Section A style of the exam.

VCE-style questions — weak vs strong responses

Five questions, **22 marks**, climbing the command-word ladder from Identify to Evaluate. Read the question, *have a real go in your head or on paper*, then reveal a weak and a strong sample answer, the marking guide, and mark yourself. Your score feeds the dock in the corner.

A **switch** connects the staff PCs to one another within the office LAN. A **router** connects the office network to the internet — that is, it links two different networks.

- 1 — switch identified as connecting devices within the LAN
- 1 — router identified as connecting the office to the internet / different networks

A firewall keeps hackers and viruses out of the network. There are different kinds you can buy.

A firewall inspects traffic at the network boundary and allows or blocks it against rules. For Clutch it could apply a rule to **deny all inbound connections to the payment server's port except from the application servers**, so strangers on the internet can't reach it. A **hardware firewall** is a dedicated device at the network edge protecting every device behind it; a **software firewall** runs on an individual computer, protecting only that machine.

- 1 — firewall allows/blocks traffic by rules at the boundary
- 1 — a valid, specific rule example (port/IP/protocol)
- 1 — hardware = dedicated edge device for whole network
- 1 — software = runs on one device, protects that device

An IDS and an IPS both detect intrusions and threats on the network to keep it safe.

When a threat is detected, an **IDS only raises an alert / log** for staff and the traffic continues, **whereas an IPS automatically blocks or drops** the traffic. An IDS typically sits **out of line**, monitoring a copy of traffic, while an IPS sits **inline** so traffic must pass through it and can be stopped in real time.

P-E-L — build a full-mark sentence

High-band responses follow **Point → Evidence → Link**: make the point, back it with a specific detail or example, then link it to the question or scenario. Write an answer to the prompt and watch the three meters fill as the keywords appear.

Command words & the traps that cost marks

Match your answer's depth to the verb. Under-answering a Compare with a one-word "Identify" answer is the most common way to leave marks behind.

Common traps & misconceptions

Exam trap

△ Exam trap — "the switch connects to the internet" No. A switch has no idea what the internet is — it only knows devices on its own network by MAC address. The **router** is what reaches other networks using IP addresses. If a question asks which component "directs traffic between the office network and the internet", the answer is the **router**.

Exam trap

△ Exam trap — "a firewall stops viruses" A firewall filters **network traffic** by rules; it is not designed to scan files for malware. If a staff member downloads an infected attachment through an allowed connection, the firewall waves it through — that's the job of **anti-malware security software**. Don't write "the firewall removes the virus." Different tools, different jobs.

Exam trap

△ Exam trap — naming the family vs the brand Write the **type of software and what it does** ("anti-malware software that scans and removes malicious programs"), not just a brand name. "We'd install [Brand]" with no explanation earns little. Examiners mark the function, not the logo.

Exam trap

△ Exam trap — using "detect" for the IPS Saying an IPS "detects threats" is only half the answer and often costs the mark. The defining word for IPS is **prevent / block**; the defining word for IDS is **detect / alert**. In a Distinguish question, your sentence must contrast those verbs: "*An IDS only alerts staff, whereas an IPS actively blocks the traffic.*"

Exam trap

△ The five KK02 traps in one place ① **Router ≠ switch** — between networks vs within a network. ② **Firewall ≠ antivirus** — filters traffic vs scans files. ③ **IDS ≠ IPS** — alert vs block. ④ **Storage ≠ backup** — live data vs a separate recoverable copy (off-site matters). ⑤ **Hardware vs software** — know which family each component belongs to, and that a software firewall straddles both.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Identify · 2 marks

Clutch's office network connects the staff PCs to one another, and also connects the office to the internet. Identify the component responsible for each, and state in a phrase what each one connects.

✗ A weak answer

er response A router and a switch. **Why it loses marks:** Both correct names, but the question also asked you to **state what each connects**. With no roles given, this caps at 1 mark.

✓ A strong answer

er response A **switch** connects the staff PCs to one another within the office LAN. A **router** connects the office network to the internet — that is, it links two different networks. **Why it scores:** Each component named *and* its connection stated. Clear 'within a network' vs 'between networks' contrast = full marks.

How the marks are awarded

- 1 — switch identified as connecting devices within the LAN
- 1 — router identified as connecting the office to the internet / different networks

Q2. Describe · 4 marks

Describe how a firewall protects Clutch's servers, including one example of a rule it might apply (). Then distinguish a hardware firewall from a software firewall (). Clutch's payment server should never accept unsolicited connections from the public internet.

✗ A weak answer

er response A firewall keeps hackers and viruses out of the network. There are different kinds you can buy. **Why it loses marks:** Vague and partly wrong — a firewall doesn't remove viruses, and 'keeps hackers out' gives no **mechanism** or rule. No real distinction between the two types.

✓ A strong answer

er response A firewall inspects traffic at the network boundary and allows or blocks it against rules. For Clutch it could apply a rule to **deny all inbound connections to the payment server's port except from the application servers**, so strangers on the internet can't reach it. A **hardware firewall** is a dedicated device at the network edge protecting every device behind it; a **software firewall** runs on an individual computer, protecting only that machine. **Why it scores:** States the allow/block mechanism, gives a concrete **rule example**, and cleanly separates edge-device vs per-machine. Full marks.

How the marks are awarded

- 1 — firewall allows/blocks traffic by rules at the boundary
- 1 — a valid, specific rule example (port/IP/protocol)
- 1 — hardware = dedicated edge device for whole network
- 1 — software = runs on one device, protects that device

Q3. Distinguish · 4 marks

Distinguish between an Intrusion Detection System (IDS) and an Intrusion Prevention System (IPS) in terms of (a) what each does when it detects a threat, and (b) where each typically sits in relation to the traffic.

✗ A weak answer

er response An IDS and an IPS both detect intrusions and threats on the network to keep it safe.

Why it loses marks: This states what they share, not how they **differ** — and a Distinguish question is marked on the contrast. It misses the block-vs-alert point entirely.

✓ A strong answer

er response When a threat is detected, an **IDS only raises an alert / log** for staff and the traffic continues, **whereas an IPS automatically blocks or drops** the traffic. An IDS typically sits **out of line**, monitoring a copy of traffic, while an IPS sits **inline** so traffic must pass through it and can be stopped in real time. **Why it scores:** Explicit contrast on both axes, using 'whereas', with the decisive alert-vs-block verbs and out-of-line vs inline placement. Full marks.

How the marks are awarded

- 1 — IDS detects and alerts only
- 1 — IPS detects and automatically blocks
- 1 — IDS out of line / monitors a copy
- 1 — IPS inline / traffic passes through it

Q4. Explain · 5 marks

Clutch stores its lesson records on a server in the office. Explain why this storage alone is not enough to protect the records, and explain how a 3-2-1 backup strategy would let Clutch recover from a ransomware attack. Ransomware encrypts every file it can reach on the network and demands payment for the key.

✗ A weak answer

er response They should back up their data so if something goes wrong they have a copy. Backups are important for safety. **Why it loses marks:** Generic. It never explains **why storage alone fails**, never defines 3-2-1, and never links to surviving ransomware. Marks for effort only.

✓ A strong answer

er response Storage holds only the live copy, so a single event — hardware failure, theft, or ransomware encrypting the files — destroys the data with no way back. A **3-2-1 strategy keeps 3 copies, on 2 different media, with 1 off-site/offline**. When ransomware encrypts everything reachable on Clutch's LAN, the **off-site/offline copy is out of its reach**, so Clutch restores the lesson records from that copy and resumes operating **without paying the ransom**. **Why it scores:** Explains the single-point-of-failure of storage, defines 3-2-1 precisely, and crucially links the **off-site copy** to ransomware recovery. Full marks.

How the marks are awarded

- 1 — storage is the live/only copy one event destroys it
- 1 — names ransomware risk (encryption of reachable files)
- 1 — 3 copies on 2 media
- 1 — 1 copy off-site/offline
- 1 — off-site copy enables recovery without paying ransom